CMPEN 331 – Lab 2 Report

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Q1. Program Output

When running the program, the following is printed in the Run I/O panel:
The sum from 0 .. 100 is :338350:
-- program is finished running --
This confirms that the program correctly calculates the sum of squares from 0 to 100.

Q2. Final Value of Register $t7

At the end of the program execution:
$t7 = 0x00002710 = 10000 (decimal)
This makes sense, since the last value of i before the loop exits is 100, and the instruction mul $t7, $t6, $t6 computes $i^2 = 100^2 = 10000$.

Q3. Registers Changed at Breakpoint

A breakpoint was set at line 13 (mul $t7, $t6, $t6).
After executing that one instruction:
- $t7 (result of multiplication)
- lo, hi (updated by multiplication)
- pc (advances to next instruction)

Across multiple iterations of the loop, additional registers change:
- $t6 (loop counter i)
- $t7 (square of i)
- $t8 (sum before update)
- $t9 (temporary sum + i^2)
- $t0 (incremented value of i)
- $a0 (used later when printing)
- pc, lo, hi (always updated per instruction)

Thus, the full list of registers that change over the course of execution is:
$t6, $t7, $t8, $t9, $t0, $a0, pc, lo, hi